

Python Beginner Programming Tasks

Topics: Variables, print(), input(), Type Conversion, Strings, Arithmetic, Comparison, Logical and Membership Operators

1. Variables + print()

1. **Student Information** — Store name, age, course, and institute in variables. Display them using print().
2. **Employee Information** — Store employee name, department, salary, and experience. Print a simple employee profile.
3. **Product Information** — Store product name, price, and quantity. Display all information.

2. input() Function

4. **Personal Introduction** — Ask the user for name, age, city, and course. Print a sentence introducing the student.
5. **Simple Greeting** — Ask for the user's name and print a welcome message.
6. **Student Registration** — Ask for name, roll number, course, and semester. Display the entered information.

3. int(), float(), str()

7. **Age Calculator** — Input birth year, convert it to int, and calculate approximate age.
8. **Shopping Bill** — Input product price and quantity. Convert price to float and quantity to int. Calculate the total bill.
9. **Temperature Converter** — Input Celsius temperature as float and convert it to Fahrenheit.
10. **Student Marks** — Input marks of three subjects, convert them to float, and calculate total and average.

4. String Concatenation

11. **Full Name Generator** — Input first name and last name and combine them using +.
12. **Email Generator** — Input first name and last name and generate an email address such as ahsan.raja@gmail.com.
13. **Address Generator** — Input house number, street, and city and combine them into one address.

5. String Repetition

14. **Pattern Generator** — Ask the user for a symbol and number, then generate a repeated pattern.

6. Arithmetic Operators

15. **Basic Calculator** — Input two numbers and calculate addition, subtraction, multiplication, division, and modulus.
16. **Rectangle Calculator** — Input length and width and calculate area and perimeter.
17. **Circle Calculator** — Input radius and calculate area and circumference.
18. **Simple Interest** — Input principal, rate, and time and calculate simple interest.
19. **Shopping Calculator** — Input price and quantity and calculate subtotal, discount, and final price.

7. Comparison Operators

20. **Age Eligibility** — Input age and check whether the person is 18 or older.
21. **Pass/Fail Checker** — Input marks and check whether marks are greater than or equal to 50.

8. Logical Operators

22. **Driving Eligibility** — Input age and whether the person has a license. Check whether age is at least 18 AND the person has a license.
23. **University Admission** — Input marks and age. Admission is allowed if marks are at least 50 AND age is at least 18.
24. **Discount Eligibility** — A customer gets a discount if purchase amount is at least 5000 OR the customer is a member.
25. **Login System** — Input username and password and check whether both are correct using the and operator.

9. Membership Operators — in / not in

26. **Character Checker** — Input a word and a character. Check whether the character exists in the word.
27. **Email Validator — Basic** — Input an email and check whether @ exists in it.
28. **Keyword Checker** — Input a sentence and check whether the word python exists in the sentence.